

Curvature Equivalences on the Elliptic Paraboloid: Extrinsic vs. Intrinsic Geometry

For any 2-dimensional Riemannian manifold, the intrinsic Ricci scalar R and the extrinsic Gaussian curvature K are explicitly linked by the identity:

$$R = 2K \tag{1}$$

Let us map out both paths step-by-step using the coordinate parameterization for the elliptic paraboloid:

$$\mathbf{x}(t, v) = \begin{pmatrix} t \\ v \\ t^2 + v^2 \end{pmatrix} \tag{2}$$

Preliminary: The First Fundamental Form (Metric Tensor)

First, we find the basis tangent vectors by differentiating $\mathbf{x}$ with respect to our coordinate parameters t and v :

$$\mathbf{x}_t = \begin{pmatrix} 1 \\ 0 \\ 2t \end{pmatrix}, \quad \mathbf{x}_v = \begin{pmatrix} 0 \\ 1 \\ 2v \end{pmatrix} \tag{3}$$

The components of our metric tensor g_{ij} are the dot products of these tangent vectors:

- $g_{tt} = \mathbf{x}_t \cdot \mathbf{x}_t = 1 + 4t^2$
- $g_{tv} = g_{vt} = \mathbf{x}_t \cdot \mathbf{x}_v = 4tv$
- $g_{vv} = \mathbf{x}_v \cdot \mathbf{x}_v = 1 + 4v^2$

The determinant of the metric, which we will call W , is:

$$\det(g) = W = (1 + 4t^2)(1 + 4v^2) - (4tv)^2 = 1 + 4t^2 + 4v^2 \tag{4}$$

Inverting the matrix g_{ij} yields the components of the inverse metric g^{ij} :

$$g^{tt} = \frac{1 + 4v^2}{W}, \quad g^{tv} = g^{vt} = \frac{-4tv}{W}, \quad g^{vv} = \frac{1 + 4t^2}{W} \tag{5}$$

1 Christoffel Symbols

To calculate curvature intrinsically, we must first determine how our coordinate basis twists across the surface using the Christoffel symbols of the second kind, given by the formula:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}) \quad (6)$$

Evaluating the metric derivatives (e.g., $\partial_t g_{tt} = 8t$, $\partial_t g_{tv} = 4v$, etc.) and contracting them with the inverse metric simplifies to a remarkably clean set of non-vanishing symbols:

For the t -coordinate ($k = t$):

$$\Gamma_{tt}^t = \frac{4t}{W}, \quad \Gamma_{tv}^t = \Gamma_{vt}^t = 0, \quad \Gamma_{vv}^t = \frac{4t}{W} \quad (7)$$

For the v -coordinate ($k = v$):

$$\Gamma_{tt}^v = \frac{4v}{W}, \quad \Gamma_{tv}^v = \Gamma_{vt}^v = 0, \quad \Gamma_{vv}^v = \frac{4v}{W} \quad (8)$$

2 The Extrinsic Way: Gaussian Curvature (K)

The extrinsic path uses the **unit normal vector field** $\mathbf{N}$ to measure how the surface bends into $\mathbb{R}^3$.

1. **Find the Unit Normal Vector:**

$$\mathbf{x}_t \times \mathbf{x}_v = \begin{pmatrix} -2t \\ -2v \\ 1 \end{pmatrix} \Rightarrow \mathbf{N} = \frac{1}{\sqrt{W}} \begin{pmatrix} -2t \\ -2v \\ 1 \end{pmatrix} \quad (9)$$

2. **Compute Second Derivatives of $\mathbf{x}$:**

$$\mathbf{x}_{tt} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}, \quad \mathbf{x}_{tv} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_{vv} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \quad (10)$$

3. **Construct the Second Fundamental Form (b_{ij}):**

Projecting these second derivatives onto our normal anchor $\mathbf{N}$ via the dot product ($b_{ij} = \mathbf{x}_{ij} \cdot \mathbf{N}$):

- $b_{tt} = \frac{2}{\sqrt{W}}$
- $b_{tv} = b_{vt} = 0$
- $b_{vv} = \frac{2}{\sqrt{W}}$

4. **Calculate Gaussian Curvature:**

Gauss defined K as the ratio of the determinants of the second and first fundamental forms:

$$K = \frac{\det(b)}{\det(g)} = \frac{\left(\frac{2}{\sqrt{W}}\right)\left(\frac{2}{\sqrt{W}}\right) - 0}{W} = \frac{4}{W^2} = \frac{4}{(1 + 4t^2 + 4v^2)^2} \quad (11)$$

3 The Intrinsic Way: Ricci Tensor and Ricci Scalar (R)

The intrinsic path ignores $\mathbf{N}$ entirely. It feeds the Christoffel symbols calculated in Section 1 straight into the Riemann Curvature Tensor:

$$R^l_{ijk} = \partial_j \Gamma^l_{ik} - \partial_k \Gamma^l_{ij} + \Gamma^l_{jm} \Gamma^m_{ik} - \Gamma^l_{km} \Gamma^m_{ij} \quad (12)$$

Step 1: Contract to the Ricci Tensor

We contract the Riemann tensor to obtain the Ricci tensor components via $\text{Ric}_{ij} = R^l_{ilj}$. Evaluating this contraction using our known fields yields:

$$\text{Ric}_{tt} = \frac{4(1+4t^2)}{W^2}, \quad \text{Ric}_{tv} = \text{Ric}_{vt} = \frac{16tv}{W^2}, \quad \text{Ric}_{vv} = \frac{4(1+4v^2)}{W^2} \quad (13)$$

Step 2: Trace to the Ricci Scalar

To get the scalar value, we contract the Ricci tensor with our inverse metric components ($R = g^{ij} \text{Ric}_{ij}$):

$$R = g^{tt} \text{Ric}_{tt} + g^{tv} \text{Ric}_{vt} + g^{vt} \text{Ric}_{tv} + g^{vv} \text{Ric}_{vv} \quad (14)$$

Plugging our inverse metric formulas into this equation:

$$R = \left(\frac{1+4v^2}{W} \right) \left(\frac{4(1+4t^2)}{W^2} \right) + 2 \left(\frac{-4tv}{W} \right) \left(\frac{16tv}{W^2} \right) + \left(\frac{1+4t^2}{W} \right) \left(\frac{4(1+4v^2)}{W^2} \right) \quad (15)$$

Factoring out the shared denominator W^3 :

$$R = \frac{4(1+4t^2+4v^2+16t^2v^2) - 128t^2v^2 + 4(1+4t^2+4v^2+16t^2v^2)}{W^3} \quad (16)$$

$$= \frac{8+32t^2+32v^2+128t^2v^2-128t^2v^2}{W^3} \quad (17)$$

$$= \frac{8(1+4t^2+4v^2)}{W^3} = \frac{8W}{W^3} = \frac{8}{W^2} \quad (18)$$

4 Conclusion

Comparing our final values from both methods:

Method	Mathematical Expression
Extrinsic (Gauss)	$K = \frac{4}{W^2}$
Intrinsic (Ricci)	$R = \frac{8}{W^2}$

The intrinsic Ricci scalar evaluates to exactly $\frac{8}{(1+4t^2+4v^2)^2}$, confirming perfectly that $R = 2K$.